Swapna K V

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SUMMARY

- Results-driven .NET Technical Lead with 15 years of experience leading software development teams and delivering scalable solutions across all SDLC phases.
- Proficient in .NET, C#, ASP.NET, SQL Server, Agile methodologies, and experienced in C++, C#, Python, and microservices architecture.
- Designed and developed cloud-based applications in Azure using .NET Core, Azure Functions, Docker, with expertise in GIT, xUnit, and Moq for testing and version control.
- Developed semiconductor inspection systems, drivers, and real-time database applications.

EXPERIENCE

Technical Lead | Photon | Ashburn | December 2022 - April 2024

- Led the design and development of API Specification and the architecture of 12 microservices using C#/.NET Core.
- Led the development of 5 **Azure Functions** and 7 containerized applications deployed to Azure Container Instances and Azure Container Registry.
- Improved system reliability by implementing unit tests with XUnit and Moq, achieving 90 % of code coverage.
- Optimized data storage with Azure CosmosDB, resulting in 25% faster data retrieval.
- Provided technical leadership, conducting code reviews and ensuring high code quality, reducing code defects by 10%.

Full Stack developer | Brewers Insight | Ashburn, VA | January 2020 - October 2022

- Contributed to the inception of Brewers Insight, defining the technical vision and technical stack for the cloud-based brewing software.
- Designed and implemented serverless APIs using API Management and Azure Functions (C#), resulting in an **80%** reduction in infrastructure costs.
- Designed and implemented an event-driven backend with Azure Functions and Event Grid, processing **10,000+ events daily.**
- Designed and implemented Azure Functions with C# and Azure Service Bus for ordered event processing.
- Designed a relational schema in Azure SQL and a non-relational schema in CosmosDB.
- Developed client applications using Angular, HTML5, and CSS and deployed using Azure Storage and Azure CDN to save up to 80 % of infrastructure cost.

Technical Architect | QuEST Global | Thiruvananthapuram, India | April 2018 - December 2019

- Led the UI development for the Component Inspection system for semiconductor chips, enabling storage and execution of inspection recipes.
- Successfully migrated the UI from VC++ to C#/WPF, reducing development time for new features by **20%** and improving application responsiveness by **15%** .
- Developed device drivers using C++ for communication with component inspection hardware.

Technical Architect | QuEST Global | Thiruvananthapuram, India | July 2006 - July 2017

- Contributed to the development and bug fixes of e-terrahabitat, a Smart Grid software solution for real-time data and process management, using C++, ASP.NET MVC, C#, and HTML.
- Led the development of the EAGLE GIS framework using WPF, C#, and SQL Server to monitor and integrate globally distributed equipment.
- Led the development of Tele-Tag, an interactive social networking gaming app for mobile devices, using ASP.NET MVC, C#, HTML5, and SQL Server, with real-time TV scene interaction and video fingerprinting.
- Led the design, development, and scheduling of a C++ I/O driver for the Bus Coupler Module (BCM), enabling communication between process hardware and control software via RS-232, TCP/IP, and USB, with OPC Server support.

Member Technical Staff | Sharp Software, | Bangalore, India | September 2005 - July 2006

- Developed and tested the Network Scanner Tool (NST) using C++ and VC++, enabling peer-to-peer scanning between networked scanners and user PCs.
- Implemented automated scanning tasks such as routing images, converting files to text, and sending files via email, along with image enhancement for better readability.
- Improved application performance through version upgrades and optimization of scanning and file processing workflows.

Software Engineer | Keane | Bangalore, India | May 2005 - September 2005

- Developed an application to convert an LDL model into corresponding Java code, utilizing C++ and Java for debugger development.
- Collaborated with cross-functional teams to ensure smooth integration of the LDL-to-Java conversion process within the broader system architecture.
- Conducted code reviews, testing, and performance optimizations to ensure reliability and efficiency in the application.

Software Engineer | NeST Technologies | Cochin, India | October 2003 - May 2005

- Developed an eReader application using the DAISY (Digital Accessible Information System) Standard, ensuring accessibility for visually impaired users.
- Built the user interface in Visual C++ (VC++), incorporating bookmarks and navigation for audiobooks.
- Implemented backend services in C++ and COM, supporting DAISY-compliant multimedia and audio synchronization.

EDUCATION

Bachelor Of Technology in Electrical and Electronics Engineering | UNIVERSITY OF KERALA

Kerala | 2001

SKILLS

Languages: C# .Net, Net Core, C++, Python, TypeScript, JavaScript

Databases: SQL Server, MySQL, CosmosDB

Tools: MS Visual Studio, VS Code, Git, JIRA

CI/CD: TeamCity, Jenkins, Azure Devops

Web Technologies: ASP .Net, ASP .Net Core, ASP .Net Web API, Angular

OS: Windows, Linux

Azure: App Services, Azure Functions, API Management, Azure Cosmos DB, Azure Container Registry, Azure Event Grid, Service Bus, Azure Container Instance, API Management